

[Unsupervised Machine Learning.]

Unsupervised machine learning is a type of ML that analyse and models data without labelled responses or predefined categories. It is basically used for the clustering.

→ {group your data into similar clusters.}

Types of algorithm.

- (i) K means clustering algorithm
- (ii) Hierarchical clustering
- (iii) DBScan clustering
- (iv) Silhouette score

→ Principal Component Analysis (PCA)

→ [Dimensionality Reduction.]

→ Features

Dataset :

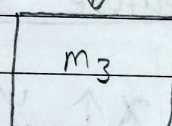
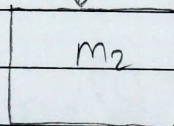
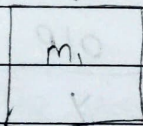
500 features.

(i) Curse of Dimensionality:

3 features

6 features

15 features



Acc 1

<

Acc 2 ↑↑

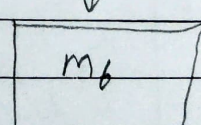
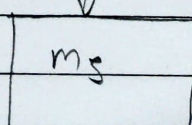
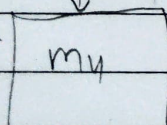
< Acc 3 ↑↑↑

>

50 features

100 features

500 features.



Acc 4 ↓↓

>

Acc 5 ↓↓↓

>

Acc 6 ↓↓↓↓

model is over fitted.

⇒ Two ways to reduce curse of dimensionality.

① Feature Selection ⇒ Imp features

② Dimensionality Reduction (PCA)

↓

Feature Extraction ⇒ Derived the new features with the help of set of features.

Feature Selection vs feature extraction.

① Why Dimensionality Reduction?

① Prevent → Curse of Dimensionality.

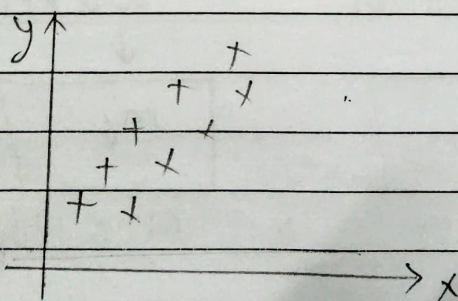
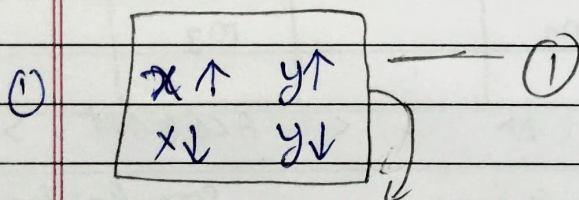
② Improve the performance of model.

③ Visualize the data → understand the data

↓

Higher dimensions [40, 50, ...]

Feature Selection.

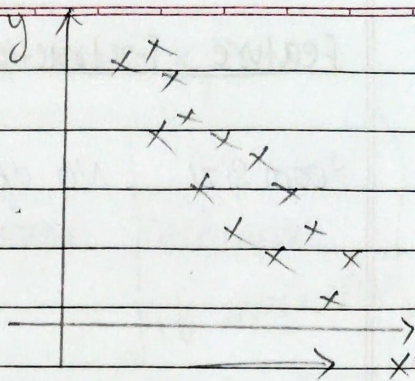


IMP	OIP
x	y
-	-
-	-
-	-

(ii)

$x \uparrow$	$y \downarrow$
$x \downarrow$	$y \uparrow$

— (ii) $\Rightarrow$



$$\text{cov}(x, y) = \frac{\sum_{i=1}^n (x_i - \bar{x}) \times (y_i - \bar{y})}{N-1}$$

(i) if $\text{cov}(x, y)$ is +ve, then we have follow the first Relation.

(ii) If $\text{cov}(x, y)$ is -ve, then we have follow the second Relation.

(iii) if $\text{cov}(x, y)$ is 0 then there is no relation between the x and y , and the plot of x & y is completely random.

$$\text{Pearson correlation} = \frac{\text{cov}(x, y)}{\sigma_x \times \sigma_y} \quad \{\text{Range} = [-1, 1]\}$$

{ The more toward the value of +1 the more +ve correlation b/w x & y is.

Feature Extraction.

Room Size	No of rooms	Price
—	—	—
—	—	—
—	—	—

Independent feature

dependent feature

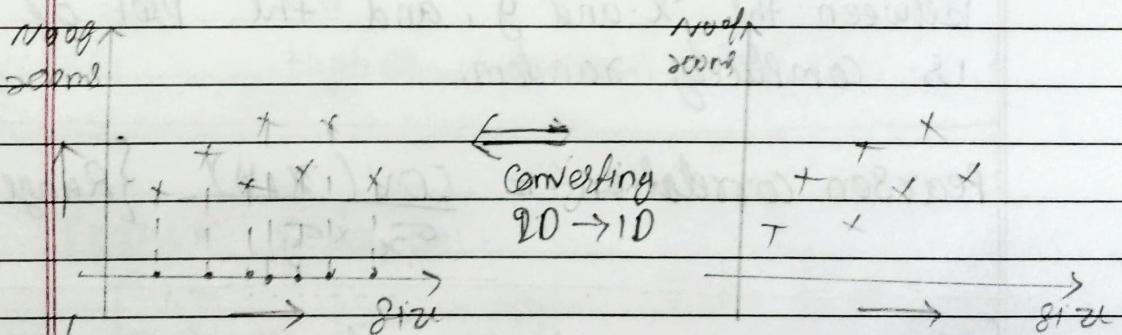
→ 2 features → 1 feature



Dimensionality Reduction.

→ apply some transformation techniques and derive a new feature "House size" and now based on this new extracted feature we are predicting the price.

PCA Geometric Intuition.



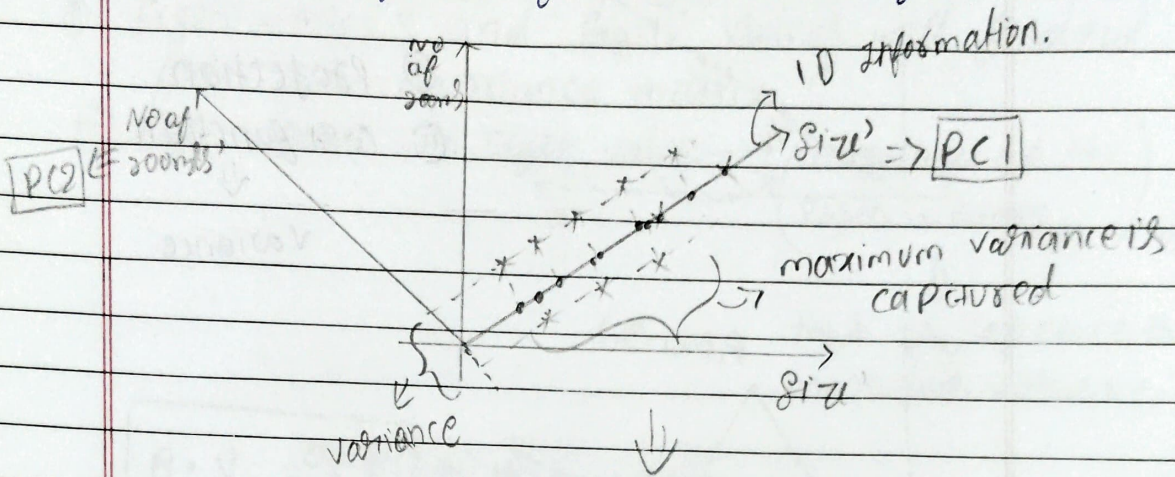
→ the No of rooms data lost when we transform the data 2D → 1D

So to reduce the data or Information loss we use the PCA

Eigen Decomposition on matrix



Kind of Transformation Techniques.



2D $\rightarrow$ 1D $\Rightarrow$ but the much Information is not lost.

- $\Rightarrow$ PC1 $\Rightarrow$ maximum variance is captured
- $\Rightarrow$ PC2 $\Rightarrow$ maximum variance is captured after PC1
- and so on if it is captured the maximum variance one-by-one

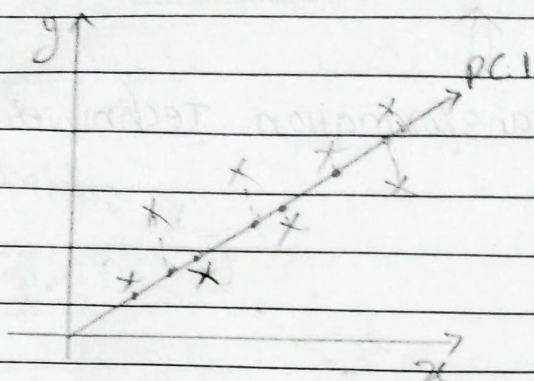
$\Rightarrow$ 2-Dimension $\Rightarrow$ we have to make PC1 and PC2

$$\Rightarrow \text{Var}(PC1) > \text{Var}(PC2)$$

$\Rightarrow$ 3-Dimension $\Rightarrow$ we have to make PC1, PC2 and PC3 for 3-Dimension.

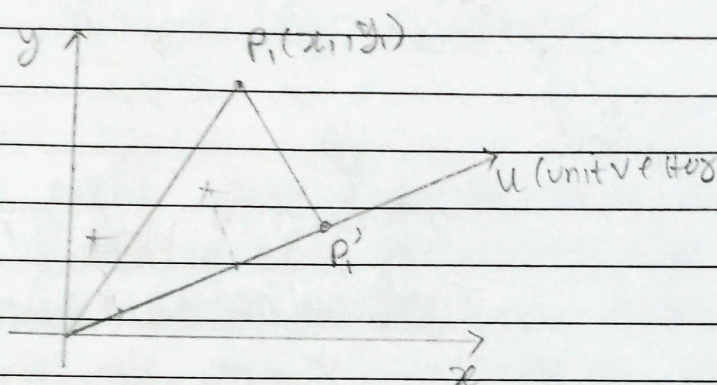
$$\Rightarrow \text{Var}(PC1) > \text{Var}(PC2) > \text{Var}(PC3)$$

maths Intuition Behind PCA Algorithm.



2D $\rightarrow$ 1D

- ① Projection
- ② cost function
 $\downarrow$
variance



$$\text{Proj}_{P_1}(u) = \frac{P_1 \cdot u}{\|u\|}$$

$\|u\| = 1$ (unit vector)

$\Downarrow$

$$\text{Proj}_{P_1}(u) = P_1 \cdot u \Rightarrow \text{scalar value.}$$

$\Downarrow$

variance.

$\Rightarrow$ Projections $\Rightarrow \{P_0', P_1', P_2', P_3' \dots P_n'\}$
of points.

$\Downarrow$

$\{x_0', x_1', x_2', x_3', \dots x_n'\}$

Cost function

$$\text{variance} = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$$

$\Rightarrow$ find the best unit vector with max variance

Eigen vectors and Eigen values.

- ① Covariance matrix between features.
- ② Eigen vectors and Eigen values will find out from this covariance matrix.
- ③ Eigen vectors $\rightarrow$ Eigen value $\rightarrow$ (magnitude of the Eigen vectors)

$\downarrow$
this is capture the maximum variance.

$$A \cdot V = \lambda V \Rightarrow \text{Characteristic Equation.}$$

$\downarrow$
Eigen vector $\rightarrow$ Has maximum magnitude

$\downarrow$
Has max eigenvalue

$\downarrow$
selected as Best Principal Component.

$\downarrow$
capture the max variance.

- ① Covariance of features.

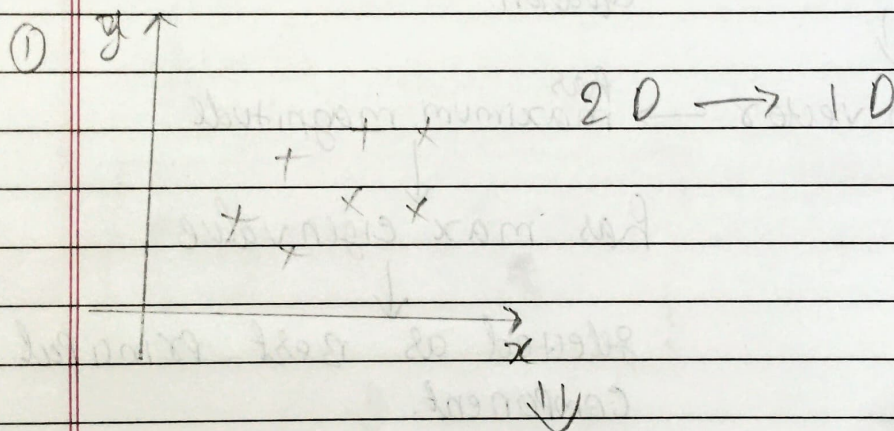
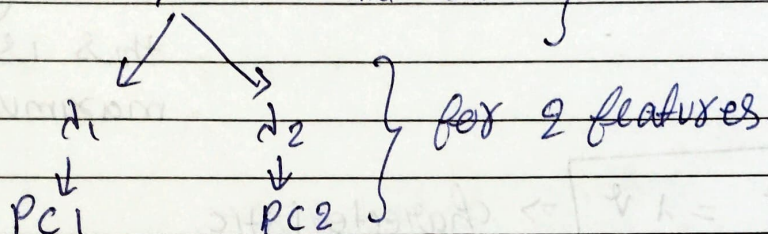
X	Y	Z
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$\downarrow$
 $X' \Rightarrow (2 \times 2)$ matrix.

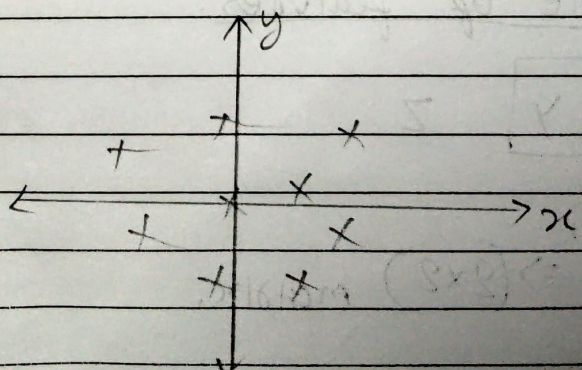
$$\text{cov}(x, y) = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{N-1}$$

$$A = \begin{matrix} & \begin{matrix} x & y \end{matrix} \\ \begin{matrix} x \\ y \end{matrix} & \begin{bmatrix} \text{var}(x) & \text{cov}(x, y) \\ \text{cov}(y, x) & \text{var}(y) \end{bmatrix} \end{matrix} \quad \begin{aligned} &\Rightarrow \text{cov}(x, x) \Rightarrow \text{var}(x) \\ &\Rightarrow \text{cov}(y, y) \Rightarrow \text{var}(y) \end{aligned}$$

$$\{ A \cdot \vec{v} = \lambda \cdot \vec{v} \rightarrow \text{unit vector} \}$$



① Standardize the data so that it will become Zero centric

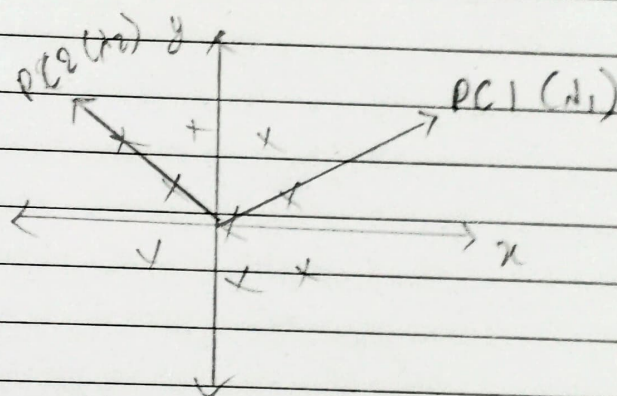
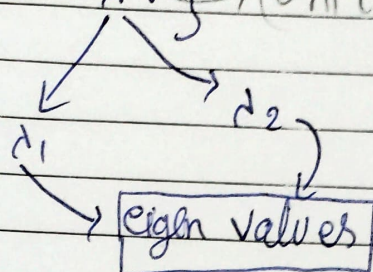


③ Covariance matrix of x & y

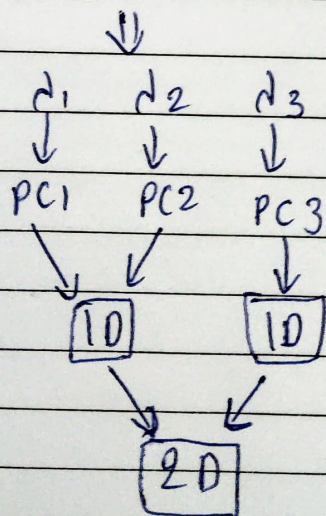
	x	y
x	$\text{var}(x)$	$\text{cov}(x, y)$
y	$\text{cov}(y, x)$	$\text{var}(y)$

④ Find out the eigen values & vectors.

$$\{A \cdot v = \lambda v\} \rightarrow (\text{unit vector})$$



3D $\rightarrow$ 2D



3D $\rightarrow$ 1D

